L02: Predicate Logic (First-order logic)

- Outline
  - Predicates
  - Quantifiers
  - Quantifiers with Restricted domains
  - Logical Equivalences involving Quantifiers
  - Negating Quantified Expressions
  - Nested Quantifiers
  - PNF

- Reading
  - Kenneth Rosen, Section 1.4-1.5

Predicate

- **Definition:** Informally, a **predicate** is a statement that may be true or false depending on the choice of values of its variables. Each choice of values produces a proposition.
- **Example:** $P(x)$ denotes the statement "x is greater than 3". x is the variable, and P is the predicate "is greater than 3". $P(4)$ is T, but $P(2)$ is F.
- More formally, a statement involving n variables $x_1, x_2, \ldots, x_n$, denoted by $P(x_1, x_2, \ldots, x_n)$, is the value of the **propositional function** P at the n-tuple $(x_1, x_2, \ldots, x_n)$ and P is called an **n-ary predicate**.

Examples

- Let $A(x)$ denote the statement "student x is required to take either COMP 2711 or COMP 2711H". Suppose Alice is a COMP student and Bob is a CHEM student. What are the truth values of $A(\text{Alice})$ and $A(\text{Bob})$?
- Let $Q(x, y)$ denote the statement "$x=y+3$". What are the truth values of the propositions $Q(1,2)$ and $Q(3,0)$?

Universal Quantification

- When the variables in a propositional function are assigned values, the resulting proposition has a certain truth value.
- Sometimes we may want to say that a predicate is true over a set of values.
- **Definition:** The **universal quantification** of $P(x)$ is the statement "for all elements x in the domain, $P(x)$ is true".
- Denote as $\forall x\ P(x)$. We read it as "for all x $P(x)$" or "for every x $P(x)$".
- Here $\forall$ is called the **universal quantifier**.
- An element for which $P(x)$ is false is called a **counterexample** of $\forall x\ P(x)$.

Domain

- The **domain** or **universe** is the set of all possible values of a variable.
- The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not well defined.
- Generally, an implicit assumption is made that the domain is nonempty. Otherwise $\forall x\, P(x)$ is true for any propositional function $P(x)$ because there are no elements x in the domain for which $P(x)$ is false.

Examples

- Let $P(x)$ be the statement "$x + 1 > x$". What is the truth value of the quantification $\forall x\, P(x)$, where the domain consists of all real numbers?
- Let $Q(x)$ be the statement "$x < 2$". What is the truth value of the quantification $\forall x\, Q(x)$, where the domain consists of all real numbers?
- Let $P(x)$ be the statement "$x^2 > 0$". What is the truth value of the quantification $\forall x\, P(x)$, where the domain consists of all real numbers?
- What is the truth value of "$\forall x\, (x^2 \geq x)$" if the domain consists of all real numbers?
  What is its truth value if the domain consists of all integers?

Existential Quantification

- **Definition:** The **existential quantification** of $P(x)$ is the statement "there exists an element x in the domain such that $P(x)$".
- The notation $\exists x\, P(x)$ denotes the existential quantification of $P(x)$.
- Here $\exists$ is called the **existential quantifier**.
- Note that $\exists x$ means "there exists **at least one** x in the domain" but not "there exists one and only one x in the domain" or "there exists a unique x in the domain".

Universal and Existential Quantifiers

- When the domain has n elements $x_1, x_2, \ldots, x_n$

  $\forall x\, P(x)$ is the same as $P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n)$,
  $\exists x\, P(x)$ is the same as $P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n)$

Universal and Existential Quantifiers

Universal and Existential Quantifiers		
Statement	When is it true?	When is it false?
$\forall x\, P(x)$	$P(x)$ is true for every x.	There exists an x for which $P(x)$ is false.
$\exists x\, P(x)$	There exists an x for which $P(x)$ is true.	$P(x)$ is false for every x.

Easy to prove cases:

- Give a counter example when $\forall x\, P(x)$ is false
- Give an example when $\exists x\, P(x)$ is true

Examples

- Let $P(x)$ be the statement "$x > 3$". What is the truth value of the quantification $\exists x\, P(x)$, where the domain consists of all real numbers?
- Let $Q(x)$ be the statement "$x = x + 1$". What is the truth value of the quantification $\exists x\, Q(x)$, where the domain consists of all real numbers?

Exercise on Predicate Logic

1. Which of the following statements (in which Z^+ stands for the positive integers and Z stands for all integers) is true and which is false? Explain why.

a) $\forall z \in Z^+ (z^2 + 6z + 10 > 20)$
b) $\forall z \in Z, (z^2 - z \geq 0)$
c) $\exists z \in Z^+, (z - z^2 > 0)$
d) $\exists z \in Z, (z^2 - z = 6)$

a) False, because if z=1, $1 + 6 \cdot 1 + 10 = 17 < 20$.
b) True. Consider the following cases:
   1) $z < 0$: true, because $z^2 - z \geq 0 \Rightarrow z^2 \geq z$. The LHS is positive, the RHS is negative.
   2) $z = 0$: true by substitution $0^2 - 0 \geq 0$.
   3) $z > 0$: true, because $z^2 - z \geq 0 \Rightarrow z^2 \geq z$, for all $z \geq 1$.
c) False because $z^2 \geq z$, , for all $z \geq 1$.
d) True, because for z=3, $3^2-3=6$ (also for z=-2).

Binding Variables

- A variable can be
  1. bound by a quantifier
     - the part of a logical expression bound by a quantifier is called its scope
  2. set to a particular value
  3. otherwise, free
- Example: $\exists x\big(P(x) \wedge Q(x)\big) \vee \forall x\, R(x) \vee Q(2) \vee S(x)$
- All variables in a propositional function must be bound or set to a particular value to turn it into a proposition
- Free variables, without a quantifier (e.g., $S(x)$) are not allowed

Precedence of Quantifiers

- The quantifiers ∀ and ∃ have higher precedence than all logical operators.
- Example
  - $\forall x\, P(x) \vee Q(x)$ is equivalent to $(\forall x\, P(x)) \vee Q(x)$
  - $(\forall x\, P(x)) \vee Q(x)$ is not a valid proposition because $Q(x)$ is not quantified
  - We should write $\forall x\, (P(x) \vee Q(x))$ or $(\forall x\, P(x)) \vee (\forall x\, Q(x))$
  - $\forall x\, (P(x) \vee Q(x))$ not equivalent to $(\forall x\, P(x)) \vee (\forall x\, Q(x))$ (next slide)
- When the scopes of two variables are disjoint we can use the same (or different) names.
  - $\exists x P(x) \vee \forall x Q(x)$ is the same as $\exists x P(x) \vee \forall y Q(y)$
  - Hint: Use different names to avoid confusion.

Example

- $P(x)$: student x has learned C++
- $Q(x)$: student x has learned Python
- Domain is all students
- Express the following statements using predicate and quantifiers:
  1. "every student has learned C++"

     $\forall x\, P(x)$
  2. "some student has learned C++"

     $\exists x\, P(x)$
  3. "every student has learned either C++ or Python"

     $\forall x\, (P(x) \vee Q(x))$
  4. "all students have learned C++, or all students have learned Python"

     $(\forall x\, P(x)) \vee (\forall y\, Q(y))$
- 3. and 4. are not the same. Example: Domain = {Alice, Bob}, Alice learned C++, Bob learned Python. $\forall x\, (P(x) \vee Q(x))$ is T and $(\forall x\, P(x)) \vee (\forall y\, Q(y))$ is F

Quantifier with Restricted Domain

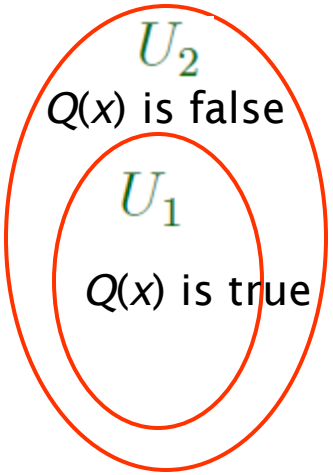
- Let U_1 and U_2 be two domains with $U_1 \subseteq U_2$. Suppose U_1 is defined as:

$$U_1 = \{\ x \in U_2 \mid Q(x) \text{ is true } \}.$$

- Then, a statement $P(x)$ about U_1 may also be interpreted as a statement about U_2

(a) $\forall x \in U_1\ (P(x))$ is equivalent to
$\forall x \in U_2\ (Q(x) \rightarrow P(x))$

(b) $\exists x \in U_1\ (P(x))$ is equivalent to
$\exists x \in U_2\ (Q(x) \wedge P(x))$



Restricted Domains

- Rewrite the following statements into equivalent statements, where the domain consists of all real numbers? (U_2 is set of real numbers)
  - (a) $\forall x < 0\ (x^2 > 0)$ (U_1 is set of negative numbers)
  - (b) $\forall y \neq 0\ (y^3 \neq 0)$ (U_1 is set of non-zero numbers)
  - (c) $\exists z > 0\ (z^2 = 2)$ (U_1 is set of positive numbers)
- **Solution**
  - (a) Equivalent to $\forall x\,(x < 0 \rightarrow x^2 > 0)$. True
  - (b) Equivalent to $\forall y\,(y \neq 0 \rightarrow y^3 \neq 0)$. True
  - (c) Equivalent to $\exists z\,(z > 0 \ \wedge\ z^2 = 2)$. True

Example on Restricted Domains 1

- Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a lion", "x is fierce", and "x drinks coffee", respectively.

  Assume that the domain consists of all creatures

- Consider the following argument:

1. Premise 1: "All lions are fierce"
2. Premise 2: "Some lions do not drink coffee"
3. Conclusion: "Some fierce creatures do not drink coffee"

- Express the statements using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.

1. $\forall x\,(P(x) \rightarrow Q(x))$
2. $\exists x\,(P(x) \wedge \neg\, R(x))$
3. $\exists x\,(Q(x) \wedge \neg\, R(x))$

- **Remark:** In the next classes, we will discuss the issue of determining whether the conclusion is a valid consequence of the premises. In this example, it is.

Example on Restricted Domains 2

- Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a professor", "x is ignorant" , and "x is vain", respectively.

  Assume that the domain consists of all people

- Express the statements using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.
- All professors are not ignorant.

  $\forall x\,(P(x) \rightarrow \neg Q(x))$
- All ignorant people are vain.

  $\forall x\,(Q(x) \rightarrow R(x))$
- All professors are not vain.

  $\forall x\,(P(x) \rightarrow \neg R(x))$
- Does the last conclusion follow the first two statements?

  No. There may well be vain professors, since the premises do not rule out the possibility that there are vain people besides the ignorant ones.

Logical Equivalence

- **Definition**

  Statements involving predicates and quantifiers are **logically equivalent** if and only if they have the same truth value no matter which predicates are substituted into these statements and which (common) domain is used for the variables in these propositional functions.

  We use the notation $S \equiv T$ to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.

Logical Equivalence Example

- Show that $\forall x\,(P(x) \wedge Q(x))$ and $\forall x P(x) \wedge \forall x Q(x)$ are logically equivalent
- **Proof**
  - Suppose $\forall x\,(P(x) \wedge Q(x))$ is true. That is, if a is in the domain, then $P(a)$ and $Q(a)$ are both true. Thus, $\forall x\, P(x)$ is true and so is $\forall x\, Q(x)$. Thus $\forall x\, P(x) \wedge \forall x\, Q(x)$ is true.
  - Suppose $\forall x\, P(x) \wedge \forall x\, Q(x)$ is true. Then, $\forall x\, P(x)$ is true and $\forall x\, Q(x)$ is true. Thus, if a is in the domain, then $P(a)$ is true and $Q(a)$ is true. Thus, for all a , $P(a) \wedge Q(a)$ is true. Thus $\forall x\,(P(x) \wedge Q(x))$ is true.

Logical Equivalence (cont'd)

- The previous logical equivalence shows that we can distribute a universal quantifier over a conjunction.

$$\forall x\,(Q(x) \wedge P(x)) \;\equiv\; \forall x\,P(x) \wedge \forall x\,Q(x)$$

- We can also distribute an existential quantifier over a disjunction:

$$\exists x\,(P(x) \vee Q(x)) \;\equiv\; \exists x\,P(x) \vee \exists x\,Q(x)$$

However, we cannot distribute a universal quantifier over a disjunction, nor can we distribute an existential quantifier over a conjunction:

$$\forall x\,(P(x) \vee Q(x)) \;\not\equiv\; \forall x\,P(x) \vee \forall x\,Q(x)$$

$$\exists x\,(P(x) \wedge Q(x)) \;\not\equiv\; \exists x\,P(x) \wedge \exists x\,Q(x)$$

Counterexample: we cannot distribute a universal quantifier over a disjunction

- Show with a counter example that $\forall x\,(P(x) \vee Q(x))$ is not equivalent to $\forall x\, P(x) \vee \forall x\, Q(x)$.
  - We already saw a counter example on Slide 14
  - Note: recall that $\forall x\, P(x) \vee \forall x\, Q(x) \equiv \forall x\, P(x) \vee \forall y\, Q(y)$
- Let $P(x)$ the predicate "x is male"
- Let $Q(y)$ the predicate "y is female"
- Let the domain be all human.
- $\forall x\,(P(x) \vee Q(x))$ means that each human is male or female (true).
- $\forall x\, P(x) \vee \forall y\, Q(y)$ means all humans are male or all humans are female (false)

General counterexample to show that we cannot distribute a universal quantifier over a disjunction

- Show that $\forall x\,(P(x) \vee Q(x))$ is not equivalent to $\forall x\,P(x) \vee \forall x\,Q(x)$.
- The domain contains only two elements a and b such that $P(a)$ is T, $Q(a)$ is F, $P(b)$ is F, $Q(b)$ is T.
- What is the truth value of $\forall x\,(P(x) \vee Q(x))$?
- True because both $P(a) \vee Q(a)$ and $P(b) \vee Q(b)$ are T.
- What is the truth value of $\forall x\,P(x)$?
- False because $P(b)$ is F
- What is the truth value of $\forall x\,Q(x)$?
- False because $Q(a)$ is F
- What is the truth value of $\forall x\,P(x) \vee \forall x\,Q(x)$?
- False.

Counterexample: we cannot distribute a existential quantifier over a conjunction

$\exists x\,(P(x) \wedge Q(x)) \not\equiv \exists x\,P(x) \wedge \exists x\,Q(x)$

Assume that the domain of x is the set of integers and the predicates $P(x)$: $2x + 1 = 5$ $Q(x)$: $x^2 = 9$

- $\exists x\,P(x) \wedge \exists x\,Q(x)$ (equivalent: $\exists x\,P(x) \wedge \exists y\,Q(y)$)
  - is true because $P(2)$ and $Q(3)$ are true.
- $\exists x\,(P(x) \wedge Q(x))$
  - is false because there is no one integer a such that $P(a)$ and $Q(a)$ are both true.
- Therefore: $\exists x\,P(x) \wedge \exists x\,Q(x)$ is NOT equivalent to $\exists x\,(P(x) \wedge Q(x))$

General counterexample to show that we cannot distribute an existential quantifier over a conjunction

- $\exists x\,(P(x) \wedge Q(x))$ is not equivalent to $\exists x\,P(x) \wedge \exists x\,Q(x)$.
- The domain contains only two elements a and b such that $P(a)$ is T, $Q(a)$ is F, $P(b)$ is F, $Q(b)$ is T.
- What is the truth value of $\exists x\,(P(x) \wedge Q(x))$?
- False because both $P(a) \wedge Q(a)$ and $P(b) \wedge Q(b)$ are F.
- What is the truth value of $\exists x\,P(x)$?
- True because $P(a)$ is T
- What is the truth value of $\exists x\,Q(x)$?
- True because $Q(b)$ is T
- What is the truth value of $\exists x\,P(x) \wedge \exists x\,Q(x)$?
- True.

Negation Example

- Let $P(x)$ denote the statement "student x has iPhone". The domain is all students in this class
- Express the following statement as a universal quantification: "every student in the class has an iPhone".
- $\forall x\ P(x)$
- Then express the negation of the statement $\neg\ \forall x\ P(x)$ using an existential Quantifier.
- "There exists (at least) a student in the class who has no iPhone"
- $\exists x\ \neg P(x)$

De Morgan's Laws for Quantifiers

De Morgan's Laws for Quantifiers			
Negation	Equivalent statement	When is it true?	When is it false?
$\neg\forall x\, P(x)$	$\exists x\, \neg P(x)$	There exists an x for which $P(x)$ is false.	$P(x)$ is true for every x.
$\neg\exists x\, Q(x)$	$\forall x\, \neg Q(x)$	$Q(x)$ is false for every x.	There exists an x for which $Q(x)$ is true.

De Morgan's Laws for Quantifiers (cont'd)

- When the domain has n elements x_1, x_2, … , x_n, it follows that

  $\neg\forall x\, P(x)$ is the same as $\neg\,(P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n))$,

  which is equivalent to $\neg P(x_1) \vee \neg P(x_2) \vee \ldots \vee \neg P(x_n)$ by De Morgan's laws. This is the same as $\exists x\, \neg P(x)$.

- Similarly, $\neg\exists x\, P(x)$ is the same as $\neg\,(P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n))$, which by De Morgan's laws is equivalent to $\neg P(x_1) \wedge \neg P(x_2) \wedge \ldots \wedge \neg P(x_n)$. This is the same as $\forall x\, \neg P(x)$.

Negation Examples

- **Example 1**

  What is the negation of the statement

  $\forall x\,(x^2 > x)$?

- Let $P(x)$ be: $x^2 > x$
- $\neg\forall x\,P(x) \equiv \exists x\,\neg P(x) \equiv \exists x\,(x^2 \leq x)$
- **Example 2**

  What is the negation of the statement

  $\exists x\,(x^2 = 2)$?

- Let $P(x)$ be: $x^2 = 2$
- $\neg\exists x\,P(x) \equiv \forall x\,\neg P(x) \equiv \forall x\,(x^2 \neq 2)$

Negation Example

- Show that $\neg\forall x\,(P(x)\rightarrow Q(x))$ and $\exists x\,(P(x) \wedge \neg Q(x))$ are logically equivalent.

- **Solution**

$$\begin{aligned}\neg\forall x\,(P(x)\rightarrow Q(x)) &\equiv \exists x \neg\,((P(x) \rightarrow Q(x))\\ &\equiv \exists x \neg\,(\neg\, P(x) \vee Q(x))\\ &\equiv \exists x\,(P(x) \wedge \neg Q(x))\end{aligned}$$

Exercise on Equivalence

- Let P(x), and Q(x) be the statements “x is a professor”, “x is ignorant”.
- "All professors are not ignorant" can be expressed as $\forall x\ (P(x) \rightarrow \neg Q(x))$.
- Which of the following statements is equivalent to $\forall x\ (P(x) \rightarrow \neg Q(x))$:
  1. $\forall x\ (\neg P(x) \vee \neg Q(x))$
  2. $\neg \exists x\ (P(x) \wedge Q(x))$
- Answer 1:Only 1
- Answer 2:Only 2
- Answer 3:Both 1. and 2.
- Correct answer: Answer 3:Both 1. and 2.
- $\forall x\ (P(x) \rightarrow \neg Q(x)) \equiv \forall x\ (\neg P(x) \vee \neg Q(x)) \equiv \forall x\ \neg(P(x) \wedge Q(x)) \equiv \neg \exists x\ (P(x) \wedge Q(x))$

Nested Quantifiers Example

- Assume that the domain for the variables x and y consists of all real numbers.

  The statement

  $$\forall x \, \forall y \, (x + y = y + x)$$

  says that $x + y = y + x$ for all real numbers x and y.

  This is the commutative law for the addition of real numbers.

Order of Quantifiers Matters

- $\forall x \exists y$ Loves(x,y) means (domain is all persons)
  - *"For every person x, there is someone y that they love" (everyone loves somebody)*
- $\exists y \forall x$ Loves(x,y) means
  - *"There is a single person y whom everyone loves"*
- $\exists x \forall y$ Loves(x,y) means
  - *"There is a person x who loves everyone".*
- $\forall y \exists x$ Loves(x,y) means
  - *"For every person y, there is someone who loves them" (everyone is loved by somebody)*
- We can switch the order of consecutive identical quantifiers: $\forall x \forall y$ Loves(x,y) $\equiv \forall y \forall x$ Loves(x,y) .

Order of Quantifiers

- $\forall x \exists y\ (x + y = 0)$ states that (domain is real numbers)
  - for every real number x , there is a real number y (additive inverse) such that $x + y = 0$, T
- What is the truth value of $\exists y \forall x\ (x + y = 0)$
  - It is false since there is no value of y that satisfies the equation $x + y = 0$ for all values of x

Example of Non Equivalence

- Let $Q(x, y, z)$ be the statement "$x + y = z$". What are the truth values of the statements

$$\forall x \, \forall y \, \exists z \, Q(x, y, z)$$

$$\exists z \, \forall x \, \forall y \, Q(x, y, z),$$

where the domain of the variables is all real numbers?

- **Solution:**
- Suppose x and y are assigned values. Then there exists a real number z such that $x + y = z$. Thus the first statement is true.
- There is no value of z that satisfies the equation $x + y = z$ for all values of x and y. Thus, the second statement is false

Math Examples

- Translate the statement "the sum of two positive integers is always positive" into a logical expression.
  - Assuming that the domain is *positive* integers:
  - $\forall x\, \forall y\, (x+y>0)$
  - Assuming that the domain is *all* integers:
  - $\forall x\, \forall y\, ((x>0 \wedge y>0) \rightarrow (x+y>0))$

- Translate the statement "every nonzero real number has a multiplicative inverse".
  - $\forall x\, ((x\neq 0) \rightarrow \exists y\, (x \cdot y = 1)) \equiv \forall x\, \exists y\, ((x\neq 0) \rightarrow (x \cdot y = 1))$
  - Note: This not always straightforward to move the quantifiers in the beginning. See discussion about PNF later

Natural Language Example 1

- Translate the statement

$$\forall x\,(C(x) \vee \exists y\,(C(y) \wedge F(x, y)))$$

into English, where $C(x)$ is "x has a computer", $F(x, y)$ is "x and y are friends", and the domain for both x and y consists of all students in the school.

- **Solution**

For every student x in the school, x has a computer or there is a student y such that y has a computer and x and y are friends.

Every student either has a computer or has a friend who has a computer.

Natural Language Example 2

- Translate the statement

$$\exists x \forall y \forall z\, ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

into English, where $F(a, b)$ means a and b are friends and the domain for x, y, and z consists of all students in the school.

- **Solution**

The statement says that there is a student x such that for all students y and all students z other than y, if x and y are friends and x and z are friends, then y and z are not friends.

In other words, there is a student none of whose friends are also friends with each other.

Natural Language Example 3

- Express the statement "if a person is female and is a parent, then this person is someone's mother" as a logical expression using the following predicates, with a domain consisting of all people.
  - $F(x)$: x is female
  - $P(x)$: x is a parent
  - $M(x, y)$: x is y's mother
- $\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y)) \equiv \forall x \exists y((F(x) \wedge P(x)) \rightarrow M(x, y))$
- Express the statement "everyone has exactly one best friend" as a logical expression using the following predicates, with a domain consisting of all people.
  - $B(x, y)$: y is x's best friend
- $\forall x \exists y((B(x,y) \wedge \forall y'((y' \neq y) \rightarrow \neg B(x,y'))))$

Negating Nested Quantifiers

- **Rule**

To negate a statement with a series of quantifiers "flip" the quantifiers and negate the remaining statement

- **Example**

Express the negation of the statement $\forall x \exists y\ (xy = 1)$ so that no negation precedes a quantifier.

$\neg \forall x\ \exists y\ (xy = 1) \equiv$

$\exists x\ \neg \exists y\ (xy = 1) \equiv$

$\exists x\ \forall y\ \neg (xy = 1) \equiv$

$\exists x\ \forall y\ (xy \neq 1)$

Quantifications of Two Variables and Negations

- The following table summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	$\neg\forall x\, \forall y\, (P(x,y)) \equiv \exists x\, \neg\forall y\, (P(x,y)) \equiv \exists x\, \exists y\, (\neg P(x,y))$
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

Quantifications of Two Variables

- The following table summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	$\neg\forall x\, \exists y\, (P(x,y)) \equiv \exists x\, \neg\exists y\, (P(x,y)) \equiv \exists x\, \forall y\, (\neg P(x,y))$
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

Quantifications of Two Variables

- The following table summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	$\neg\exists x\, \forall y\, (P(x,y)) \equiv \forall x\, \neg\forall y\, (P(x,y)) \equiv \forall x\, \exists y\, (\neg P(x,y))$
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

Quantifications of Two Variables

- The following table summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

$\neg\exists x\, \exists y\, (P(x,y)) \equiv \forall x\, \neg\exists y\, (P(x,y)) \equiv \forall x\, \forall y\, (\neg P(x,y))$

Quantifications of Two Variables

- The following table summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	$P(x,y)$ is false for every pair x, y.

Natural Language Example and Negation

- Use quantifiers to express the statement "there is a woman who has taken a flight on every airline in the world".
- Let $P(w,f)$ be "woman w has taken flight f"
- Let $Q(f,a)$ be "f is a flight on airline a"

Solution: $\exists w\, \forall a\, \exists f\, ((P(w, f) \wedge Q(f, a))$

- **Negation**

Use quantifiers to express the statement "there does not exist a woman who has taken a flight on every airline in the world" so that no negation precedes a quantifier.

$\neg\, \exists w\, \forall a\, \exists f\, ((P(w,f) \wedge Q(f,a)) \equiv \forall w\, \neg \forall a\, \exists f\, ((P(w,f) \wedge Q(f,a)) \equiv \forall w\, \exists a\, \neg\, \exists f\, ((P(w,f) \wedge Q(f,a)) \equiv \forall w\, \exists a \forall f\, (\neg(P(w,f) \wedge Q(f,a)) \equiv$

$\forall w\, \exists a\, \forall f\, (\neg P(w,f) \vee \neg Q(f,a)) \equiv \forall w\, \exists a\, \forall f\, (P(w,f) \rightarrow \neg Q(f,a))$

For every woman, there is an airline such that for all flights, this woman has not taken that flight or the flight does not belong to the airline (alternatively, if the woman took the flight, then the flight does not belong to the airline).

Null Quantifications: How can we change the scope of quantifiers?

Rules: If variable x does not appear in A (A is a logical expression bound by a quantifier whose scope does not cover x, e.g., $\forall y\, P(y)$)

Rule 1.

$$\forall x\ (A \wedge P(x)) \equiv A \wedge \forall x\ P(x) \equiv \forall x\ P(x) \wedge A \equiv \forall x\ (P(x) \wedge A)$$

Rule 2.

$$\forall x\ (A \vee P(x)) \equiv A \vee \forall x\ P(x) \equiv \forall x\ P(x) \vee A \equiv \forall x\ (P(x) \vee A)$$

Rule 3.

$$\exists x\ (A \wedge P(x)) \equiv A \wedge \exists x\ P(x) \equiv \exists x\ P(x) \wedge A \equiv \exists x\ (P(x) \wedge A)$$

Rule 4.

$$\exists x\ (A \vee P(x)) \equiv A \vee \exists x\ P(x) \equiv \exists x\ P(x) \vee A \equiv \exists x\ (P(x) \vee A)$$

Proof or rules

- Proof: Let the domain of x be $\{x_1, \ldots, x_n\}$
- Rule 1: $\forall x\ (A \wedge P(x)) \equiv A \wedge \forall x\ P(x)$

$\forall x\ (A \wedge P(x)) \equiv (A \wedge P(x_1)) \wedge \cdots \wedge (A \wedge P(x_n)) \equiv$
$A \wedge P(x_1) \wedge \cdots \wedge P(x_n) \equiv A \wedge \forall x\ P(x)$

- Rule 2: $\forall x\ (A \vee P(x)) \equiv A \vee \forall x\ P(x)$

$\forall x\ (A \vee P(x)) \equiv (A \vee P(x_1)) \wedge \cdots \wedge (A \vee P(x_n)) \equiv$
$A \vee (P(x_1) \wedge \cdots \wedge P(x_n)) \equiv A \vee \forall x\ P(x)$

- Rule 3: $\exists x\ (A \wedge P(x)) \equiv A \wedge \exists x\ P(x)$

$\exists x\ (A \wedge P(x)) \equiv (A \wedge P(x_1)) \vee \cdots \vee (A \wedge P(x_n)) \equiv$
$A \wedge (P(x_1) \vee \cdots \vee P(x_n)) \equiv A \wedge \exists x\ P(x)$

- Rule 4: $\exists x\ (A \vee P(x)) \equiv A \vee \exists x\ P(x)$

$\exists x\ (A \vee P(x)) \equiv (A \vee P(x_1)) \vee \cdots \vee (A \vee P(x_n)) \equiv$
$(A \vee ((P(x_1) \vee \cdots \vee P(x_n)) \equiv A \vee \exists x\ P(x)$

- Note that all rules hold also for $\neg P(x)$
  - E.g., Rule 1: $\forall x\ (A \wedge \neg P(x)) \equiv A \wedge \forall x\ \neg P(x)$

PNF: Prenex Normal Form

- PNF means that all the quantifiers are in the beginning
- Every formula in predicate logic can be written in PNF
- When I convert to PNF, I can use Rules 1-4, provided that all quantifiers are not negated, and all operators are ∨ or ∧.
- Conversion Steps

1. Use different variable names
   - ∃x P(x) → ∀**x** Q(**x**) should change to ∃x P(x) → ∀**y** Q(**y**)
2. If there are operators other than ∨ or ∧, replace them with equivalent expressions that contain only ∨ or ∧
   - ∃x P(x) → ∀y Q(y) should change to ¬∃x P(x) ∨ ∀**y** Q(**y**)
3. Make sure that none of the quantifiers is negated
   - ¬∃x P(x) ∨ ∀**y** Q(**y**) should change to ∀x ¬P(x) ∨ ∀**y** Q(**y**)

- Now you are ready to apply Rules 1-4
  - If you extend the scope of ∀x, and then ∀y you get ∀x∀y (¬P(x) ∨ Q(y))
  - If you extend the scope of ∀y, and then ∀x you get ∀y∀x (¬P(x) ∨ Q(y))
- ∃xP(x)→∀yQ(y) ≡ ∀x∀y(¬P(x)∨Q(y)) ≡ ∀x∀y(P(x)→Q(y)) ≡ ∀y∀x(P(x)→Q(y))

PNF Exercise

Convert $\exists z(\exists y\ R(y,z) \rightarrow \forall y\ Q(y,z))$ to PNF.

$\exists z(\exists y\ R(y,z) \rightarrow \forall y\ Q(y,z)) \equiv$ variable change from y to x

$\exists z(\exists y\ R(y,z) \rightarrow \forall x\ Q(x,z)) \equiv$ implication equivalence

$\exists z(\neg\exists y\ R(y,z) \vee \forall x\ Q(x,z)) \equiv$ De Morgan for $\neg\exists y\ R(y,z)$

$\exists z(\forall y\ \neg R(y,z) \vee \forall x\ Q(x,z)) \equiv$ Rule 2 on scope of $\forall x$ (*A* is $\forall y\ \neg R(y,z)$)

$\exists z\forall x(\forall y\ \neg R(y,z) \vee Q(x,z)) \equiv$ Rule 2 on scope of $\forall y$ (*A* is $Q(x,z)$)

$\exists z\forall x\forall y(\neg R(y,z) \vee Q(x,z)) \equiv$ already PNF, implication equivalence

$\exists z\forall x\forall y(R(y,z) \rightarrow Q(x,z))$

- I can reorder quantifiers of the same type
- $\exists z\forall x\forall y(R(y,z) \rightarrow Q(x,z)) \equiv \exists z\forall y\forall x(R(y,z) \rightarrow Q(x,z))$
  - In this example it depends on which scope, $\forall x$ or $\forall y$, I extend first

Example 1: Is $\exists x\ P(x) \rightarrow \exists y\ Q(y)$ equivalent to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$ (already PNF)?

- $\exists x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \exists x\ P(x) \vee \exists y\ Q(y)$

Domain	a	b
P	T	F
Q	F	F

Consider the domains:

Both $\neg\ \exists x\ P(x)$ and $\exists y\ Q(y)$ are false.

Therefore, $\exists x P(x) \rightarrow \exists y\ Q(y) \equiv \neg \exists x P(x) \vee \exists y\ Q(y) \equiv$ F

- $\exists x \exists y\ (P(x) \rightarrow\ Q(y)) \equiv \exists x \exists y\ (\neg\ P(x) \vee Q(y))$

Using the above domains we have True for pairs $(P(b), Q(a))$, $(P(b), Q(b))$ because $\neg\ P(b)$ is T.

- Explanation of non equivalence using PNF of $\exists x\ P(x) \rightarrow \exists y\ Q(y)$:

$\exists x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \exists x\ P(x) \vee \exists y\ Q(y) \equiv$

$\forall x\ \neg P(x) \vee \exists y\ Q(y) \equiv$ by rule 2 $\forall x\ (\neg P(x) \vee \exists y\ Q(y)) \equiv$ by rule 4

$\forall x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv \forall x \exists y\ (P(x) \rightarrow\ Q(y))$

- $\forall x \exists y\ (P(x) \rightarrow\ Q(y))$ not equivalent to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$
- Although the expressions seem similar in their original form, they are different in PNF

Example 2: Is $\forall x \exists y\, (P(x) \rightarrow Q(y))$ equivalent to $\exists y \forall x (P(x) \rightarrow Q(y))$?

- Recall from the previous slide that $\exists x\, P(x) \rightarrow \exists y\, Q(y) \equiv \forall x \exists y\, (P(x) \rightarrow Q(y))$ in PNF. Is it also equivalent to $\exists y \forall x\, (P(x) \rightarrow Q(y))$?
  - I will apply the reverse transformations
- LHS: $\forall x \exists y\, (P(x) \rightarrow Q(y)) \equiv \forall x \exists y\, (\neg P(x) \vee Q(y)) \equiv \forall x\, (\neg P(x) \vee \exists y\, Q(y)) \equiv \forall x \neg P(x) \vee \exists y\, Q(y) \equiv \neg \exists x P(x) \vee \exists y\, Q(y) \equiv \exists x P(x) \rightarrow \exists y Q(y)$
- RHS: $\exists y \forall x\, (P(x) \rightarrow Q(y)) \equiv \exists y \forall x\, (\neg P(x) \vee Q(y)) \equiv \exists y\, (\forall x \neg P(x) \vee Q(y)) \equiv \forall x \neg P(x) \vee \exists y Q(y) \equiv \neg \exists x P(x) \vee \exists y Q(y) \equiv \exists x P(x) \rightarrow \exists y Q(y)$
- Thus: $\forall x \exists y\, (P(x) \rightarrow Q(y)) \equiv \exists y \forall x\, (P(x) \rightarrow Q(y)) \equiv \exists x P(x) \rightarrow \exists y Q(y)$
  - In the previous slide I can obtain $\exists y \forall x\, (P(x) \rightarrow Q(y))$, instead of $\forall x \exists y\, (P(x) \rightarrow Q(y))$, if I change the order in which I apply rules 2 and 4.
- This is an exception: usually when changing the order of different quantifiers leads to non-equivalent expressions

Example 3: Is $\forall x\ P(x) \rightarrow \exists y\ Q(y)$ equivalent to $\forall x \exists y\ (P(x) \rightarrow\ Q(y))$?

- $\forall x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \forall x\ P(x) \vee \exists y\ Q(y) \equiv$ DeMorgan

  $\exists x\ \neg P(x) \vee \exists y\ Q(y)\ \equiv$(applying Rule 4 to the scope of x)

  $\exists x\ (\neg P(x) \vee \exists y\ Q(y)) \equiv$(applying Rule 4 to the scope of y)

  $\exists x \exists y\ (\neg P(x) \vee Q(y)) \equiv \exists x \exists y\ (P(x) \rightarrow\ Q(y))$
- Thus: $\forall x\ P(x) \rightarrow \exists y\ Q(y) \equiv \exists x \exists y\ (P(x) \rightarrow\ Q(y))$
- In slide 51 we established that
  - $\forall x \exists y\ (P(x) \rightarrow\ Q(y))$ $(\equiv \exists x\ P(x) \rightarrow \exists y\ Q(y))$ is NOT equivalent to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$ $(\equiv \forall x\ P(x) \rightarrow \exists y\ Q(y))$
  - Thus: $\forall x\ P(x) \rightarrow \exists y\ Q(y)$ is NOT equivalent to $\forall x \exists y\ (P(x) \rightarrow\ Q(y))$